It can further be observed that the operational costs for one million input tokens range from 0.040¢ for Llama 3.2 1B on GPU to 6.282¢ for Gemma 2 9B on the RPi 5. Even in the most expensive scenario, this remains 2.39 times cheaper than the service cost of the OpenAI API, while in the best-case scenario, it is 375 times lower. For one million output tokens, the energy costs span from 1.014¢ for Qwen 2 0.5B on the Orin CPU up to 32.906¢ for Gemma 2 9B on the RPi 5. Even in the worst-case scenario, the cost remains 1.82 times lower than the OpenAI API service cost for GPT-4o mini, while in the best-case scenario, it is 59.17 times lower.

However, several caveats should be considered when interpreting these cost estimates. First, we do not account for power supply inefficiencies or energy usage during loading and idle states. Second, the operational costs presented here only consider energy consumption, whereas other expenses, such as device maintenance, may also contribute to the total cost. Third, these estimates do not factor in model quality, meaning that higher cloud service prices may be justified by more sophisticated model capabilities. Thus, to derive definitive conclusions, more precise energy measurements and a more comprehensive cost model incorporating additional operational factors are required.

Despite these considerations, the significantly lower operational costs—particularly for smaller models on GPU—suggest that self-deploying LMs at the edge could be more cost-efficient than cloud services in certain scenarios. This is especially relevant when idle edge resources can be leveraged, as this eliminates device acquisition costs. Moreover, self-deployment allows greater flexibility in model selection, enabling the deployment of the smallest model that meets task-specific requirements. Given that smaller models consume substantially less energy, as demonstrated in our results, this approach could yield further cost benefits. However, in scenarios where highly sophisticated model capabilities are essential, cloud services may still be the more economical option.

Beyond comparing edge inference with cloud services, it is also valuable to analyze the cost differences between the tested devices. As shown in Figure 17, GPU inference on the Orin incurs significantly lower energy costs than CPU inference on the RPi 5. However, the initial device cost must also be considered. While the RPi 5 is much more affordable, the Orin's higher efficiency raises the question of when its purchase becomes justifiable from a cost perspective. Based on retail prices from DigiKey's US store at the time of writing, the RPi 5 (including power supply and fan) costs $97, whereas the Orin is priced at $598.80—resulting in a price difference of $501.80.

To better understand the cost-effectiveness of choosing the Orin over the RPi 5, we calculated break-even times for 12 scenarios with varying input/output token ratios and device utilization (μ). The break-even time represents the duration needed for the operational cost savings of the Orin to offset its higher purchase price. For these calculations, we assumed an electricity price of 20¢/kWh. The RPi 5 was tested with the default governor (*ondemand*) and throughput-optimal thread counts, while the Orin used the default power mode (*15W*). Since *Q4_0* quantization proved more energy-efficient across both inference phases, we based our comparison on its results, excluding the energy consumption of the loading phase. As shown in Figure 18, input-heavy workloads result in shorter break-even times than output-heavy workloads, as the energy savings of GPU inference are more pronounced in the prefill phase than in the generation phase. Even with $\mu = 100\%$ utilization (i.e., continuous inference with no idle time), the fastest break-even scenario takes nearly 9 years. For balanced or output-heavy workloads, the required time increases even further, making it unlikely that the Orin would ever financially break even in typical edge AI deployments.

Model	I/O-ratio=25/75				I/O-ratio=50/50				I/O-ratio=75/25			
	u=25%	u=50%	u=75%	u=100%	u=25%	u=50%	u=75%	u=100%	u=25%	u=50%	u=75%	u=100%
G2-9B	39.50	19.75	13.17	9.88	37.85	18.92	12.62	9.46	34.75	17.38	11.58	8.69
L3.1-8B	38.91	19.45	12.97	9.73	38.01	19.00	12.67	9.50	36.34	18.17	12.11	9.08
I2.5-7B	39.48	19.74	13.16	9.87	38.84	19.42	12.95	9.71	37.61	18.81	12.54	9.40
Y1.5-6B	40.12	20.06	13.37	10.03	36.39	18.20	12.13	9.10	29.87	14.94	9.96	7.47
P3.5-M	43.84	21.92	14.61	10.96	39.41	19.70	13.14	9.85	31.34	15.67	10.45	7.84
L3.2-3B	43.43	21.72	14.48	10.86	39.08	19.54	13.03	9.77	31.00	15.50	10.33	7.75
G2-2B	55.57	27.79	18.52	13.89	50.89	25.45	16.96	12.72	41.58	20.79	13.86	10.39
Q2-1.5B	63.80	31.90	21.27	15.95	58.21	29.10	19.40	14.55	47.64	23.82	15.88	11.91
L3.2-1B	54.81	27.41	18.27	13.70	50.80	25.40	16.93	12.70	42.86	21.43	14.29	10.71
FT5-L	70.07	35.03	23.36	17.52	70.45	35.22	23.48	17.61	71.45	35.72	23.82	17.86
Q2-0.5B	146.09	73.05	48.70	36.52	139.04	69.52	46.35	34.76	124.23	62.11	41.41	31.06

Fig. 18. Break-even time in years between RPi 5 and Orin GPU for different IO-token ratios and device utilizations.

While these break-even time estimates provide insight, several caveats must be considered when interpreting the results. First, electricity prices have a direct inverse impact on break-even times. Doubling the electricity price would halve the time required to offset the Orin's higher purchase cost. Second, computational performance is not accounted for in this analysis. In this respect, the higher throughput of the Orin may justify its higher purchase price for certain computationally demanding use cases. The utilization values in Figure 18 are based on the Orin, which is the more powerful device. As a result, in some scenarios, the RPi 5 would require utilization exceeding 100% to match the Orin's performance, which is not feasible. Consequently, the break-even times for workloads corresponding to $\mu = 100\%$ on the RPi 5 are significantly longer than the minimum values observed. Thus, for workloads where the RPi 5 provides sufficient performance, it remains the more cost-effective choice over the Orin.

Main Takeaways

- Edge inference could potentially yield cost benefits over cloud services due to its flexibility in model choice.
- From a pure cost perspective the RPi 5 is the more efficient device than the Orin.

7 Conclusions

In this paper, we explored the feasibility and trade-offs of LM inference on edge devices, evaluating key performance factors such as memory usage, inference speed, and energy consumption across CPU- and GPU-based platforms. Our findings highlight the complex interplay between model size, hardware constraints, and quantization strategies, emphasizing the need for careful optimization to enable practical deployment in edge environments. While challenges remain with the current state-of-the-art technologies, our results also reveal promising opportunities, particularly in specific usability scenarios where local inference enhances responsiveness and autonomy, as well as potential cost benefits over cloud-based alternatives. We are optimistic that current trends, such as the emergence of GenAI-optimized Neural Processing Units (NPUs) [20, 48], will further improve the state-of-the-art of LM inference at the edge, and that this study can contribute to advancing the R&D landscape of LM inference-aware hardware optimizations, hardware-aware

model optimizations, improved inference frameworks, and adaptive resource management, ultimately making edge-centric AI systems more efficient and scalable.

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